PHILIPPINE SOCIETY OF
ANIMAL NUTRITIONISTS

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Fueling Progress: Shaping Animal Nutrition
for a Competitive and Sustainable Future

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Okada Manila









PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
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PHILIPPINE DAIRY INDUSTRY

Current Status and Innovations in Ruminant Feeding & Nutrition

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Outline of the Presentation

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Industry Overview

02
Key Players of the Industry

03
Key Legislative and Policy Support

04
Major Government Investments & Interventions

05
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1. Industry Overview

Import Dependence

98%

of dairy needs are sourced abroad

Value of Dairy Production

₱1.75B

in 2025

Growth in Dairy Production Value

10.8%

increase per year

Expansion Potential

High

Rising demand for dairy products and nutrition programs creates huge space for local dairy expansion.

Philippine Ruminant Population and Dairy Animal Inventory, as of Jan. 2026

CATTLE

2.51M
total population

41,631 (1.7%)
dairy animals

86%
smallhold

CARABAO

2.60M
total population

70,758 (2.7%)
dairy animals

99%
smallhold

GOAT

3.56M
total population

41,193 (1.2%)
dairy animals

99%
smallhold

SHEEP

79.5T
total population

Philippine ruminant production remains predominantly smallholder-based, while dairy animals form a very small portion of the national ruminant inventory.

Source: PSA

2. Key Players: A Smallholder-Based Dairy System

KEY PLAYERS

Individual Smallholder Producers

- Primary source of local milk production
- Milk output depends on management, nutrition, genetics, and animal health
- Represent 99.2% of dairy cattle farmers

Cooperatives

- Pool milk and connect farmers to markets
- Support processing and value-added dairy products
- Link farmers to programs, partnerships, and milk feeding initiatives

Dairy Processors

- Convert raw milk into marketable dairy products
- Ensure quality, safety, and shelf life
- Support value-adding and product diversification

Government Institutions/Agencies

- Policy direction and dairy program coordination
- Herd build-up, genetic improvement, milk feeding, and milk safety programs
- Research, extension, training, and farmer capacity building

Private Organizations

- Invest in production, processing, and distribution
- Support product development and market expansion
- Help scale the industry through technology and partnerships

3. Key Legislative and Policy Support

Republic Acts and policy mandates supporting the Philippine dairy industry

RA 7307 | PCC Creation
Created the Philippine Carabao Center to develop the buffalo for meat, **milk**, and hide.

RA 7884 | NDA Creation
Created the National Dairy Authority to accelerate the development of the Philippine **dairy industry**.

RA 10611 | Food Safety Act
Strengthens the **food regulatory system** to ensure that all food sold is safe to eat.

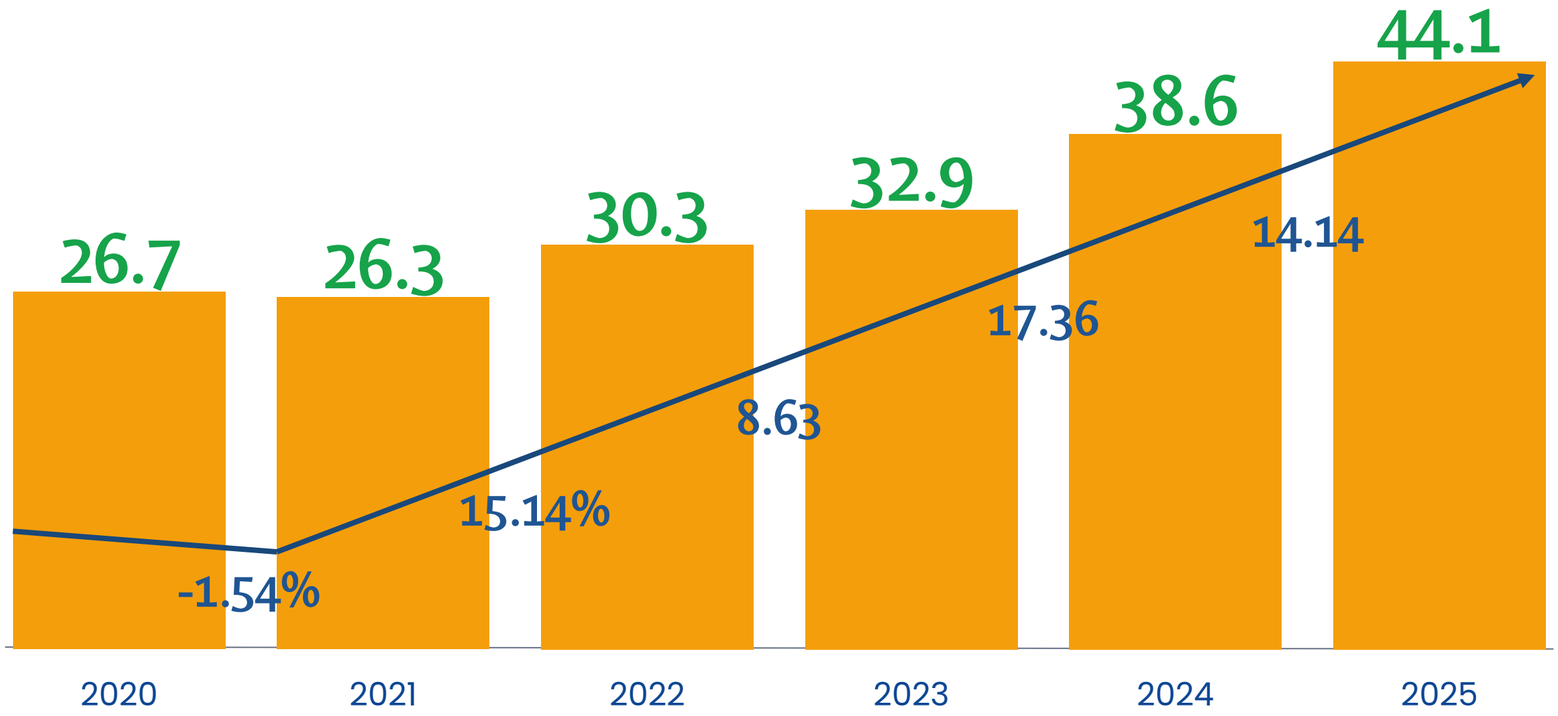
RA 11037 | Masustansyang Pagkain para sa Batang Filipino
Implements the **national milk feeding program** for undernourished children using locally produced milk.

RA 11321 | Sagip Saka Act
Aims to boost the income and livelihoods of local farmers and fisherfolk through **direct market access** and institutional support.

RA 11524 | Coconut Farmers and Industry Trust Fund
Complements NDA **dairy farming** programs as a viable livelihood and enterprise diversification for coconut farmers by providing dairy cattle, production, and management support.

RA 12308 | Animal Industry Development Act
Aims to modernize the livestock, poultry, and **dairy sectors** by providing **₱20 billion in funds annually for 10 years.**.

Volume of milk production in M liters (2020-2026)



26.7M L

2020 local milk production

44.1M L

2025 local milk production

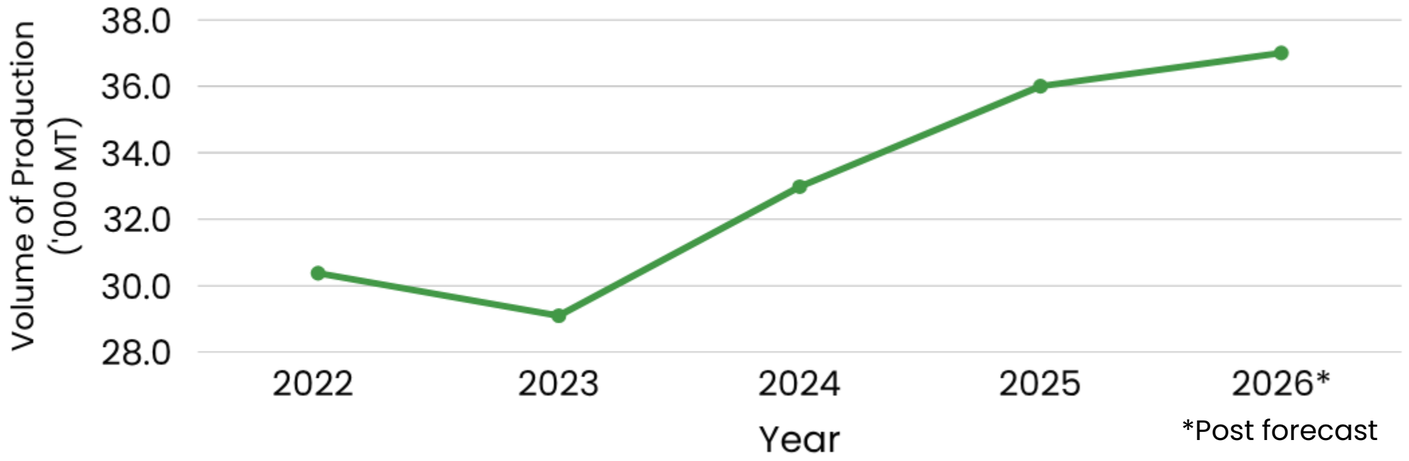
Local milk production has grown steadily from 2020-2025.

Sources: PSA OpenSTAT, and NDA-DMS

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Dairy Supply & Demand (In '000 MT, in LME)

Domestic production is rising



17.16 L → 19.53 L

reported per-capita consumption, 2025 to 2026 forecast

2025 consumption is still import-dependent



Source: NDA

What is driving local dairy growth?

Growth is being pulled by demand and pushed by herd capacity.

The Philippine dairy industry steadily expanded during the last five years.

44.0M liters produced in 2025

Significant growth from under 20M liters recorded in 2016.

Key drivers

01 **Public & Private Investments**
Funding from government programs and private enterprises.

02 **Milk Feeding Program**
Structured demand that strengthens local milk buying channels.

03 **Herd Growth**
Expansion and multiplication of dairy animals nationwide.

Contributors: Cattle + Goat Industry | Carabao Industry



4. Major Government Investments

Infrastructure

5

Dairy Stock Farms

Infrastructure to support national production capacity

Herd Expansion

1,328 heads

2025-2026

Importation of additional high-quality dairy stock to rapidly accelerate genetic and herd growth.

Program Funding

₱3.9B

2025-2026

Dedicated milk feeding program funds are allocated to secure local demand and improve child nutrition.

Breeding & Reproduction

Sexed semen Facility at PCC

Buffalo & Cattle

Advanced facility for producing more female calves for faster herd upgrading

Sources: NDA and PCC Report 2026

DA-NDA stock farms are distributed across major island groups







Sources: NDA and PCC Report 2026

NDA and PCC Program Budget Allocations, 2020-2026

Year	Budget Allocation (in '000 PHP)	
	NDA	PCC
2020	109,476	635,739
2021	647,215	621,769
2022	932,584	1,220,623
2023	834,286	1,282,379
2024	386,142	1,021,671
2025	549,967	883,237
2026*	2,549,820	2,092,672

*As of June 2026

Programs:

- Dairy Herd Build-Up
- Dairy Enterprise Development
- Coconut Farmers Industry Fund
- Milk Feeding Subsidy
- Locally Funded Projects
- US PL480 Program Funded Project
- National Livestock Program
- Operationalization of Sexed Semen Facility

Sources: NDA Budget Unit Fund Status Monitoring Report & PCC Budget Section Report

Milk Feeding Program

- Strengthens demand for locally produced milk
- Supports the livelihood and income of local dairy farmers
- Helps address child malnutrition and undernutrition
- Supports physical growth, cognitive development, and school performance
- Promotes regular milk consumption among young children



Source: NDA Report 2025

Milk Feeding Program

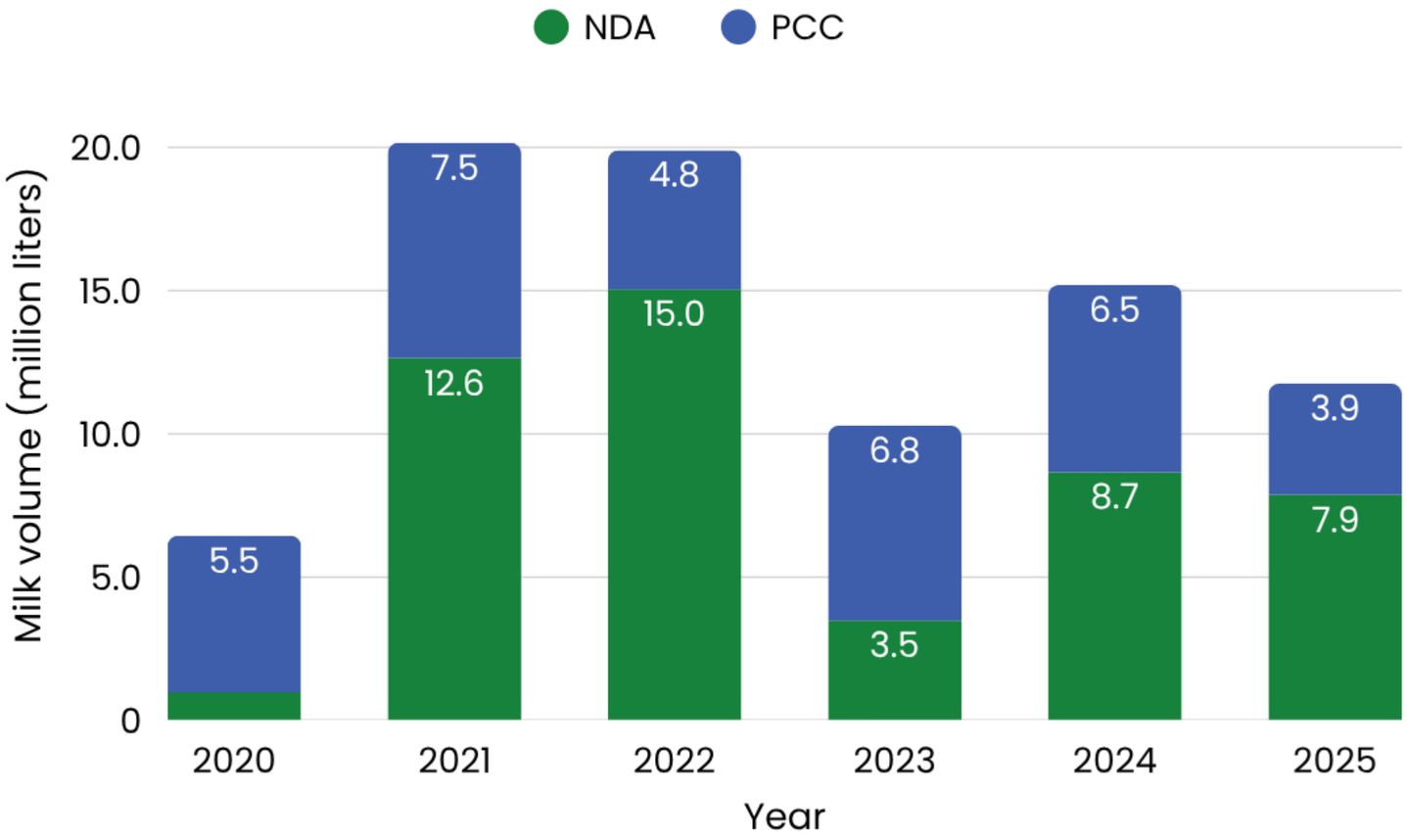
Milk Volume

83.7M L
milk served nationwide

Agency contribution

NDA	**48.68M L**	58.2%
PCC	**35.03M L**	41.8%

Peak volume: 20.15M liters in 2021.



Source: NDA Report 2025

Milk Feeding Program

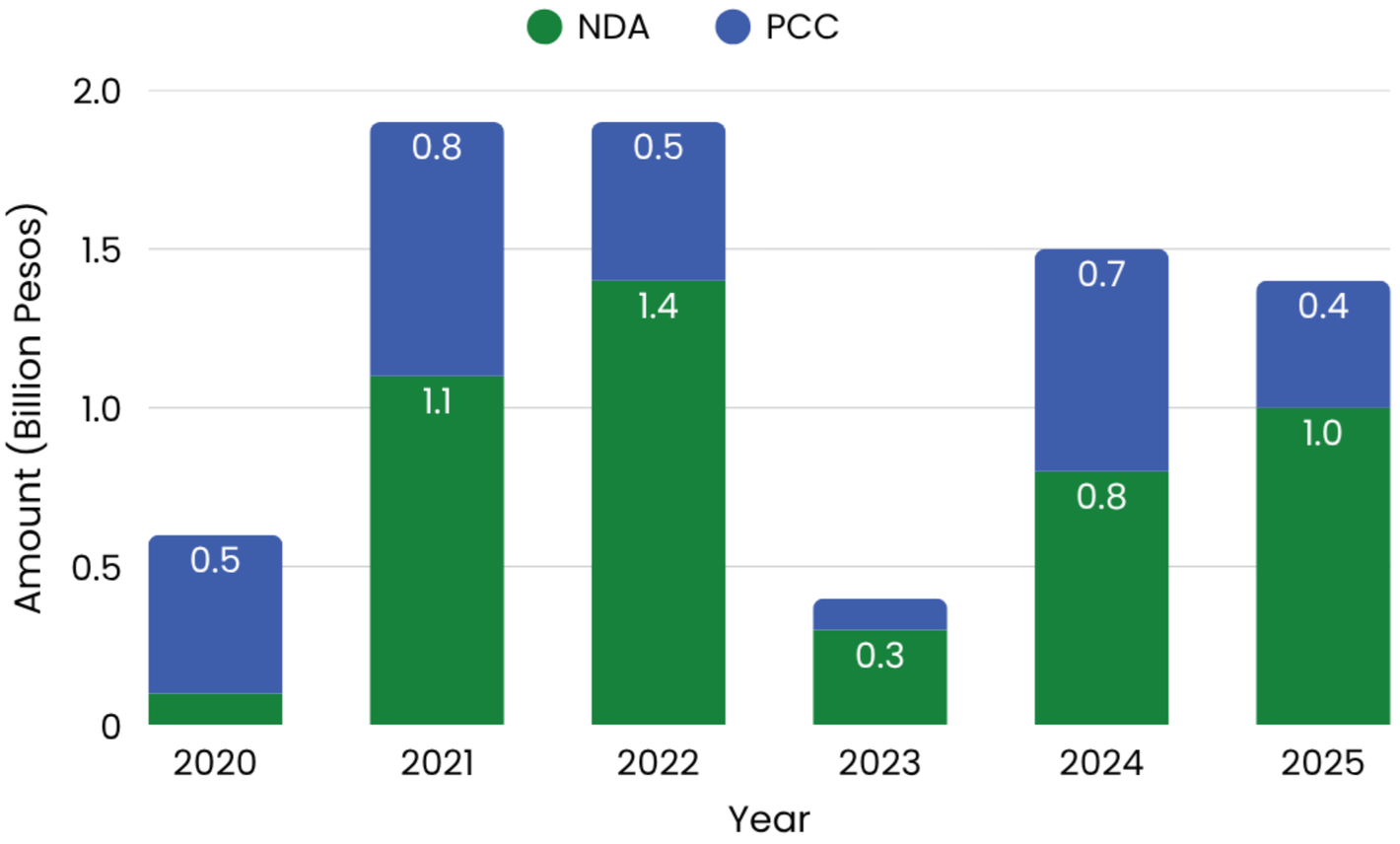
Milk Sales

₱7.78B
recorded milk sales

Agency contribution

NDA	**₱4.79B**	61.6%
PCC	**₱2.99B**	38.4%

Highest annual sales: ₱1.95B in 2022.



Source: NDA Report 2025

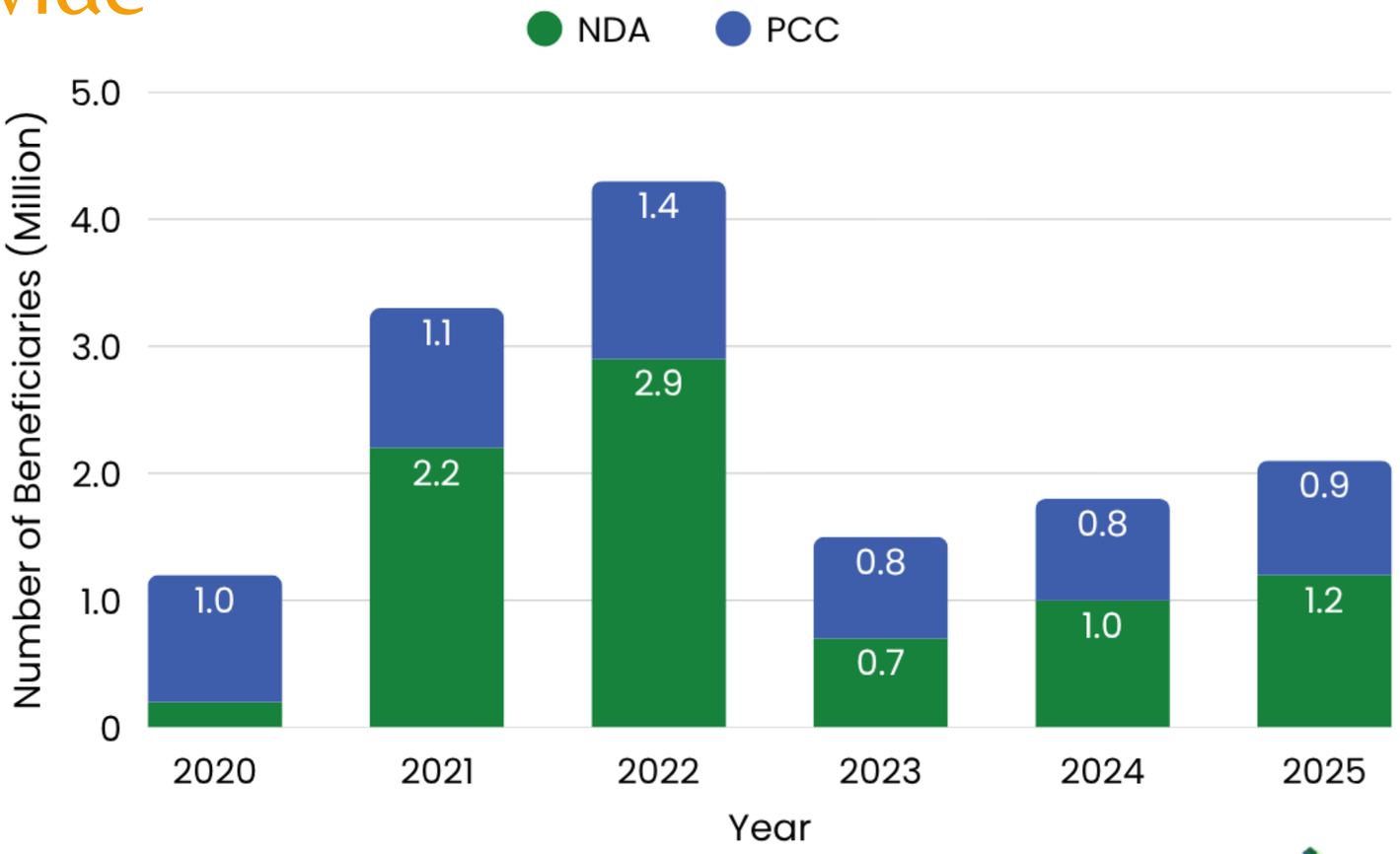
Milk Feeding Program
Beneficiaries Nationwide

14.28M
beneficiaries served

Agency contribution

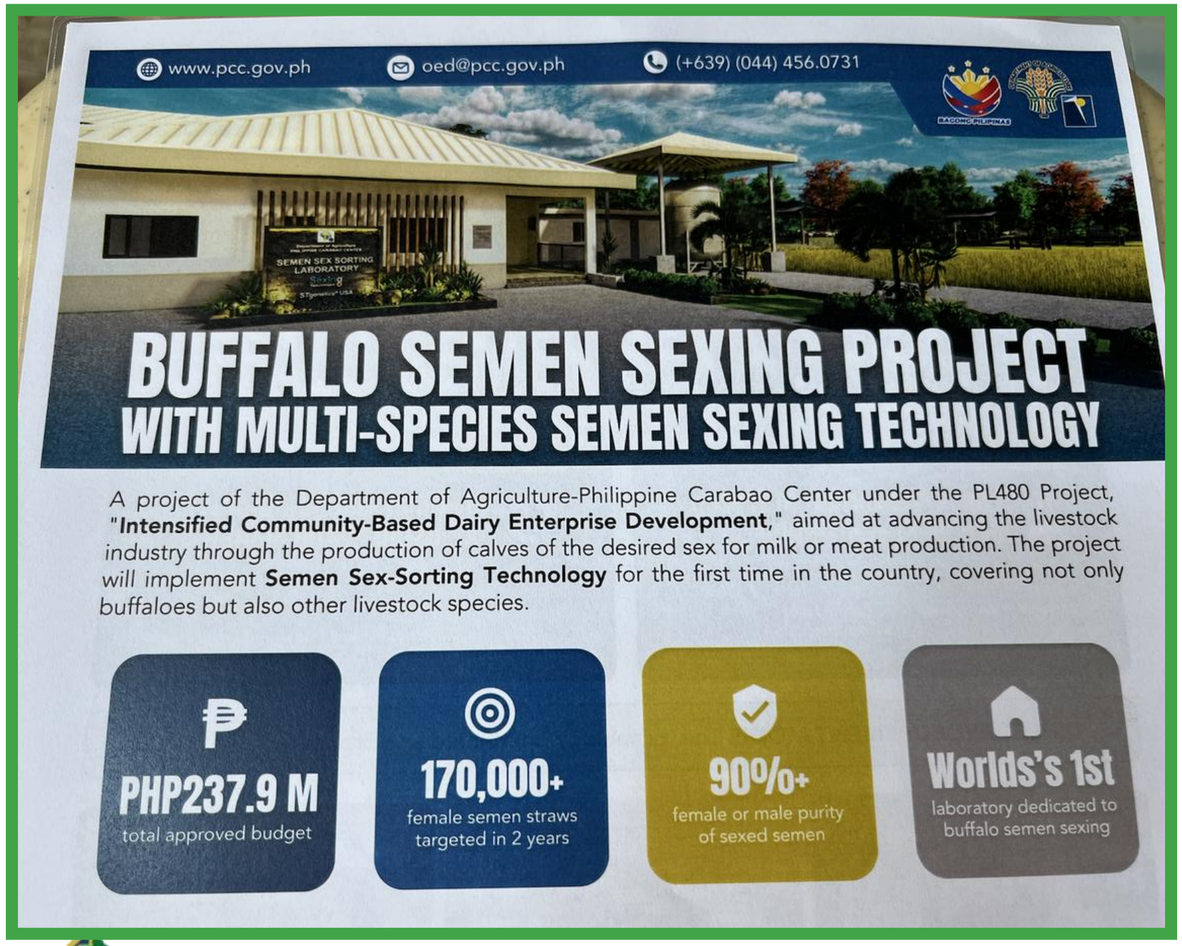
NDA	8.19M	57.3%
PCC	6.09M	42.7%

Widest reach: 4.27M beneficiaries in 2022.



Source: NDA Report 2025

Semen Sex-Sorting Facility

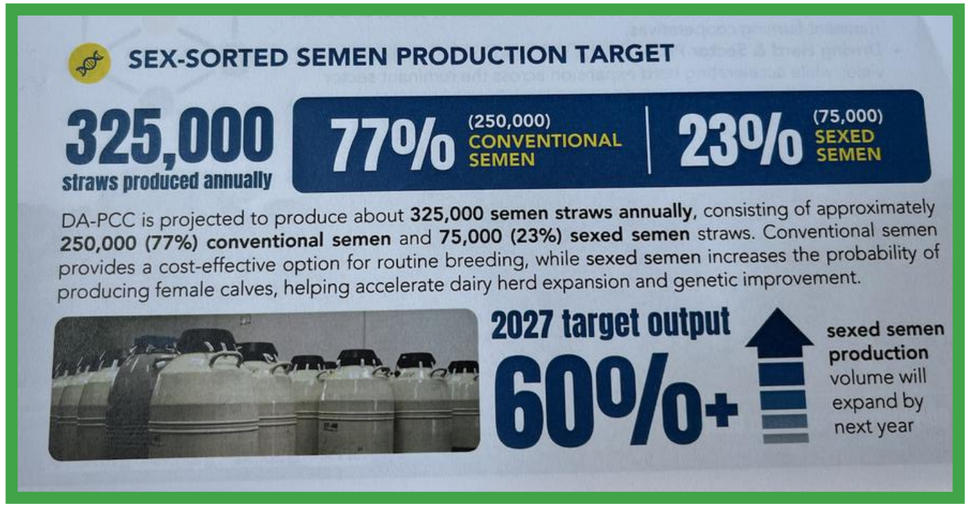


Purpose

Increases the probability of producing female calves

Industry implication

Provides more replacement heifers, supporting faster herd expansion



5. Innovations in Ruminant Feeding Systems

Smallhold System

Grass-based feeding with limited supplementation (due to high cost)

Semi & Commercial System

Corn silage & grass-based feeding system with concentrate mix as major supplement

FEED AVAILABILITY & RESOURCE DYNAMICS

Commercial Availability

Supplements, especially concentrate feeds are available commercially

Major Challenge

Roughages (grass, corn/sorghum silage) supply remains to be a **BIG challenge**

Five approaches to strengthen the feed resources

01 SCALE UP
Corn silage production
Mechanized silage production using 5–10% of total corn-planted areas

02 DIVERSIFY
Sweet sorghum for silage
Promote sweet sorghum as an alternative silage crop

03 INNOVATE
Total Mixed Ration silage
Develop and utilize TMR silage for complete-feed preservation

04 UPGRADE
Enriched rice straw
Improve the feeding value of rice straw through enrichment

05 DEVELOP
Cassava foliage silage
Use cassava foliage as an alternative protein-rich feed resource

From seasonal forage to a more resilient feed supply

MORE VOLUME	NEW CROPS	BETTER VALUE
Corn silage mechanization	Sweet sorghum + TMR silage	Rice straw + cassava foliage

Goal: strengthen year-round feed availability for dairy production

Mechanization makes large-scale forage conservation possible





Why it matters

Mechanization reduces labor constraints and enables larger volumes of feed to be harvested and conserved ahead of periods of feed scarcity.







Expanded corn silage production

Best crops for silage

- **Corn fodder :**
  - High-yielding quality crop
  - Consistent source of forage
  - Highly digestible and palatable feed
  - Higher sugar (8-9% TSS) content than grass 4-5% TSS

- **Corn silage**
  - High-energy forage for high-producing herds
    - 65 to 70% TDN vs 55 to 58% for Napier grass
  - Less labor to produce a ton of biomass
    - Yield = 40-45 tons/ha (low-density planting)
    - Yield – 50-55 tons/ha at high-density planting





Sweet Sorghum for Silage



Why it matters

High DM yield • ~14% sugar • Crude protein content of 7.9% • Supports higher feed intake

Harvest performance

Age at harvest (days)	Herbage yield (t/ha)	Dry matter (%)
75	65	23.4
80	64	28.8
85	**78**	**31.1**
90	80	33.2

Recommended: 85 days
78 t/ha and 31.1% DM

Source: PCC Report

Effects of Feeding Silage to Dairy Buffaloes

Corn

Milk yield
kg/lactation

1,992

Feed Consume, kg/d

Silage	7.41
Conc.	3.00
Total	**10.41**
%BW	2.30

Digestibility, %

OM	**76.17[a]**
DM	**66.71[a]**
CP	**73.20[a]**

Highest milk yield

Sweet Sorghum

Milk yield
kg/lactation

1,773

Feed Consume, kg/d

Silage	7.88
Conc.	3.00
Total	**10.88**
%BW	2.38

Digestibility, %

OM	**70.54[b]**
DM	**62.32[b]**
CP	**61.09[b]**

Alternative forage option

Napier

Milk yield
kg/lactation

1,560

Feed Consume, kg/d

Silage	9.95
Conc.	3.00
Total	**12.95**
%BW	2.79

Digestibility, %

OM	**71.70[b]**
DM	**63.11[b]**
CP	**60.38[b]**

Highest total intake

Source: PCC Data

Enriched Rice Straw (e-RS) Silage

Upgrading a local by-product into valuable ruminant feedstock



Collect mannually or mechanically

→



Enhance

→



Feed

Environmental gains

- Reduces carbon footprint
- Less environmental pollution
- Reduces greenhouse gas emission

Improved nutritive value

- CP: 4.9% → 8.9%
- CP digestibility: 55% → 69.2%
- DM digestibility: 49.2% → 56.1%

Total Mixed Ration Silage

Precision rationing combined with natural preservation

Integrated formulation

Balanced ration formulation preserved as silage.

Extended shelf life

Converts high-moisture by-products into stable reserves.

TMR SILAGE

Consistent nutrition

Reduces selective feeding and supports stable intake.

Labor efficiency

Supports bulk processing and streamlined feeding.

Main value: consistent feed quality with longer storage life and easier distribution.

Cassava Foliage Silage



Fresh foliage



Chopped foliage



Ensiled foliage

319,500 ha
cassava area planted in 2026

Main producing areas: Northern Mindanao, Central Luzon, and MIMAROPA

24-30%
dry matter range

17-21%
crude protein range

30-49%
fiber component

53-74%
effective degradability

Cassava Top Silage Supplementation

Item	0	0.75	1.50	2.25	SEM	*p*-value
Milk yield, kg/day	12.7[a]	13.2[a]	13.3[a]	**14.0[b]**	0.93	0.04
3.5% FCM, kg/day	14.6[a]	14.9[a]	16.1[b]	**17.2[c]**	1.01	0.03
Fat, %	4.4	4.3	4.8	**5.0**	0.35	0.58
Protein, %	3.2	3.5	3.6	**3.8**	0.39	0.34
Lactose, %	4.4	4.4	4.3	**4.5**	0.14	0.75
Solids-not-fat, %	9.3	9.3	9.2	**9.0**	0.55	0.97

Key response at 2.25 kg/day DM

Milk yield increased to 14.0 kg/day and 3.5% FCM to 17.2 kg/day.

Milk composition remained broadly comparable across inclusion levels.

Source: Wanapat et al. (2016)

6. Takeaway Messages

Industry Support

The growing ruminant and dairy industry is backed by supportive laws and funding.

Socio-Economic Impact

Continued public-private investment drives improvements in farmers' livelihoods and combats child malnutrition.

Ruminant Nutrition

Enhanced productivity relies on appropriate ruminant feeding.

Thank you for your attention

Questions and discussion

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